Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2021

Course: B.Tech
Code: 104504

Subject: LINEAR INTEGRATED CIRCUITS AND APPLICATIONS

Time: 3 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are EIGHT questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Answer any seven of the following questions :** **[2 x 7 = 14]**

(a) What is a comparator?
(b) Define settling time of D/A converter.
(c) How does a precision rectifier differ from the conventional rectifier?
(d) What output voltage would be produced by a D/A converter whose output range is 0 to 10 V and whose input binary number is 0101 for a 4-bit DAC?
(e) What is the main drawback of dual slope ADC?
(f) Give the classification of voltage regulators with example.
(g) A PLL frequency multiplier has an input frequency of f and a decade counter is included in the loop. What will be the frequency of the PLL output?
(h) Why is frequency compensation required in operational amplifiers?
(i) Draw a sample and hold circuit.
(j) Which is fastest ADC? Give proper reasoning.

Q.2 **Solve both questions :** **[7 x 2 = 14]**

(a) With the help of neat circuit diagram, explain the working of a basic bandgap voltage reference.
(b) Explain the steps used for fabrication of integrated diodes with necessary diagrams.

Q.3 **Solve both questions :** **[7 x 2 = 14]**

(a) Sketch the basic circuit using OPAMP to perform the mathematical operation of differentiation and explain its working. What are the limitations of the basic differentiator?
(b) Define output offset voltage. Explain the methods to nullify offset voltage.

Q.4 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the working of multiplier using suitable OPAMP circuit diagram and supporting equations.
(b) Draw the circuit diagram of a phase locked loop (PLL) used as amplitude modulation (AM) detector and explain its operation.

Q.5 **Solve both questions :** **[7 x 2 = 14]**

(a) Draw the block diagram and explain principle of working, characteristic and applications of a frequency synthesizer.
(b) Explain the working of a switched capacitor filter with supporting circuit diagram. How can a resistor be realized using switched capacitor?

Q.6 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the working of weighted resistor D/A converter with circuit diagram.
(b) With a suitable diagram, explain the working of successive approximation type A/D converter.

Q.7 **Solve both questions :** **[7 x 2 = 14]**

(a) Discuss the principle of operation of NE565 PLL circuit.

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(b) If an IC 555 timer is used to generate a ramp voltage with constant collector current of 1 mA, V_{CC} = 15 V and C = 0.1 μF, what would be the slope of the ramp generated? How much is the peak value of the ramp generated? What is the duration of the ramp?

Q.8 **Solve both questions :** **[7 x 2 = 14]**

(a) Design a timer which should turn on heater immediately after pressing a push button and should hold heater in ON state for 6 seconds.

(b) Explain the monostable operation of 555 timer.

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